EN 580.133

Intro to RNA-based Therapeutics

Jan 6, 2025

Created by Elaine Zhelan Chen

Objectives

1. Student will learn the compositions and structure of mRNA, siRNA, and ASOs
2. Student will learn the modes of action of various types of RNA-based therapeutics with a focus on mRNA, siRNA, and ASOs.
3. Student will learn of preclinical and clinically-approved RNA-based therapeutics examples.
4. Student will gain a fundamental understanding of challenges in RNA-based therapeutics delivery and immune toxicity
5. Student will learn various strategies for RNA-therapeutic delivery
6. Student will learn current solutions for lowering immune toxicity
7. Students will gain a fundamental understanding of the current efforts in RNA-based therapeutics research and potentials of RNA-based therapeutics

About this course

- Grade breakdown
  - Problem set #1 = 20%
  - Problem set #2 = 20%
  - Problem set #3 = 20%
  - Group presentation = 35% (30% presentation + 5% teammate evaluation)
  - Attendance = 5% (1% per class)
    - Extra 1% per class after 5 classes

- Final presentation
  - On Wednesday, Jan 15, 2025
  - **Select topic before lecture on Wednesday, Jan 8, 2025**
  - Will go over the topics towards the end of this lecture

About this course

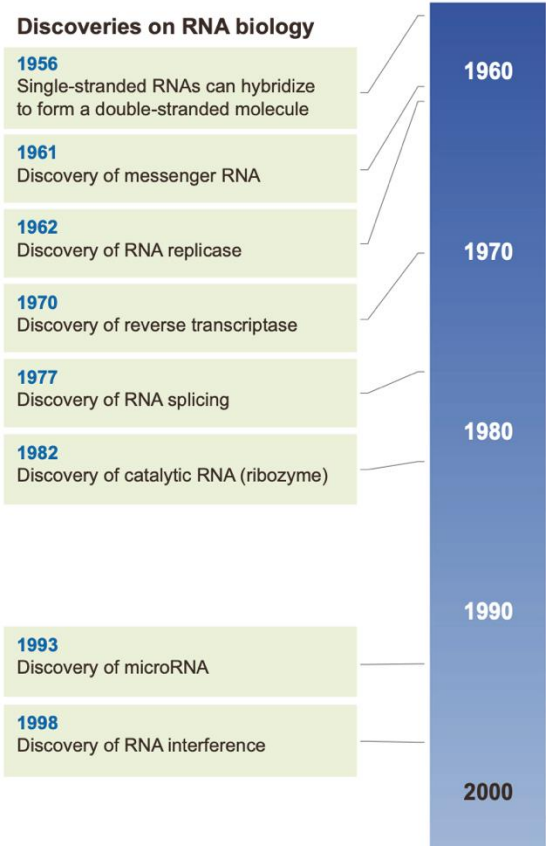
- Elaine Chen
- zchen119@jhu.edu

- Office hour by appointment only
- For emails, I will reply
  - on the day for emails received by 10PM
  - on the 2nd day for emails received after 10PM
  - Please include keywords like “EN.580.133” or “RNA therapeutics” in the subject

RNA therapeutics

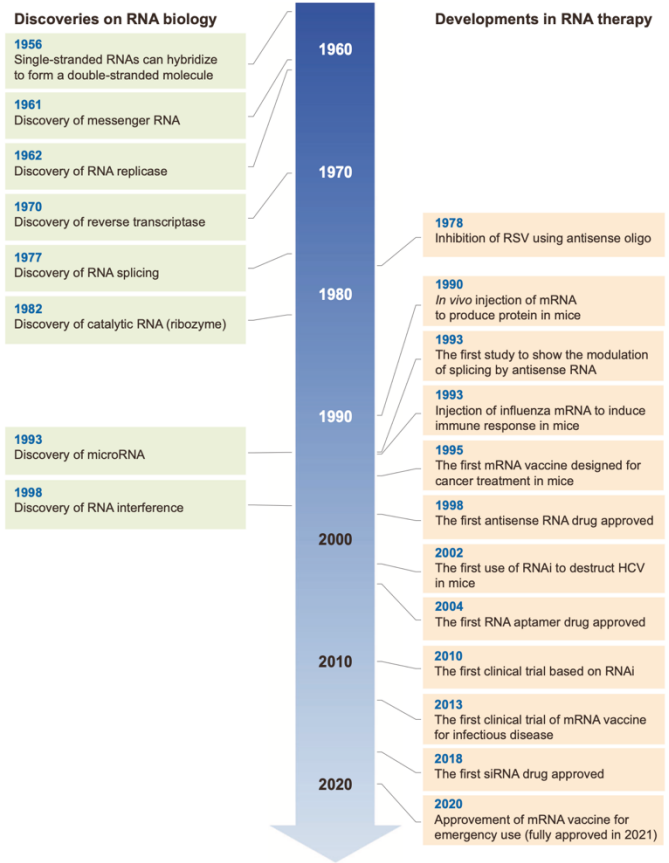
Key discoveries in RNA biology

- First described in 1958 as a key player in the flow of genetic information by Crick in his study "Central Dogma of Molecular Biology"
- Confirmed by the discovery of mRNA in 1961
  - Key role in the translation of genetic information
- Base pairing of RNA demonstrated by Rich and Davies in 1956
- Discovery of microRNA in 1993
- Discovery of RNA interference in 1998



Key milestones in RNA therapeutics

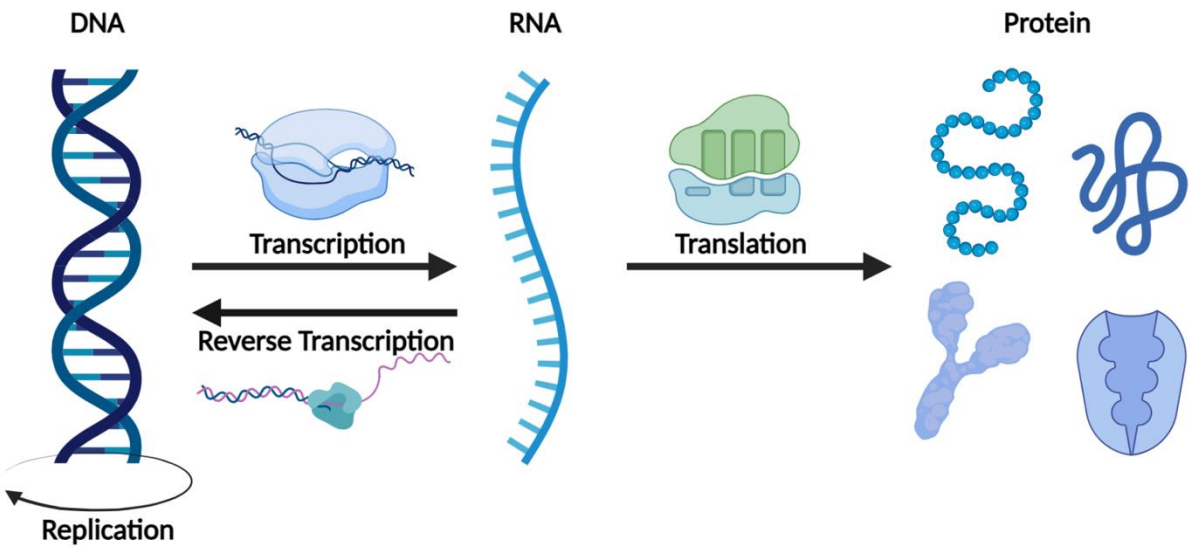
- First ever RNA-based drug approved was the antisense drug in 1998
- First Aptamer drug approved in 2004
- First siRNA drug approved in 2018
- First mRNA vaccine approved for emergency use in 2020



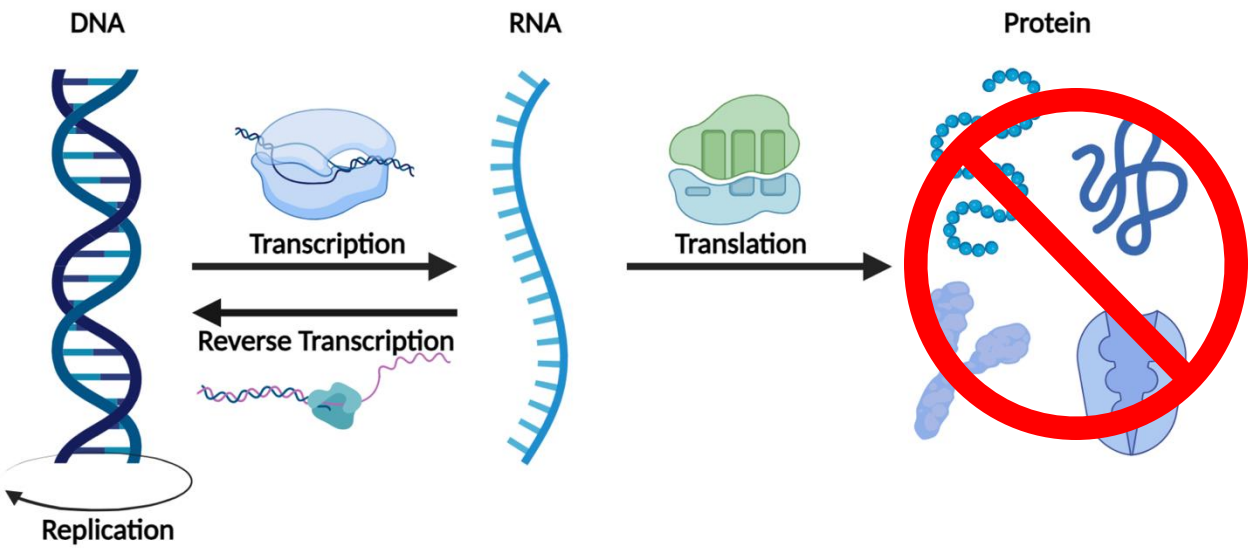
Advantages of RNA as therapeutics

1. <u>Targeting undruggable targets</u>
   - The ability to target any genetic components within the cell
   - Potential solution for undruggable targets using small molecules and biologics

Central dogma and the cause of diseases

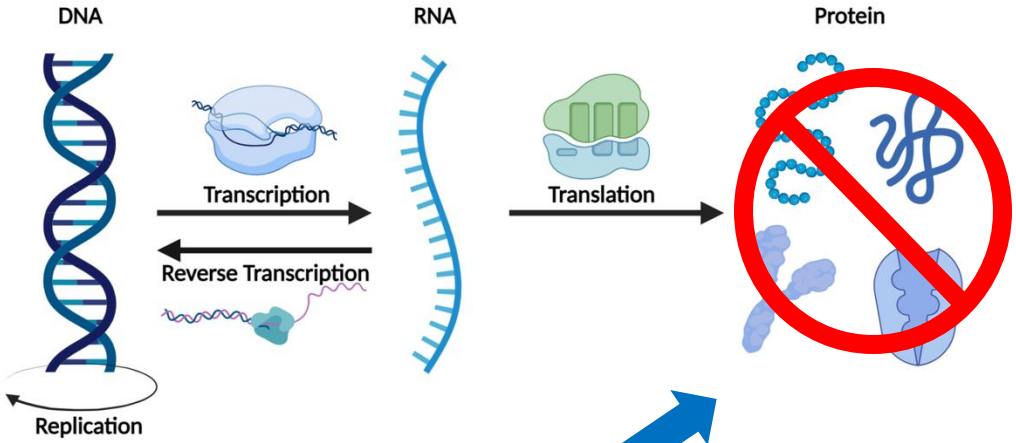


Central dogma and the cause of diseases



↑ Upregulation
↓ Downregulation
╳Misfolding/dysfunctional mutation

Small molecules as drugs



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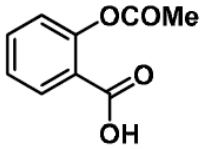
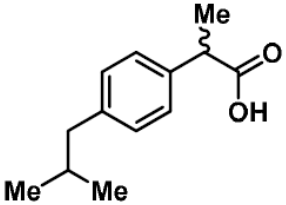
Small molecule drugs



Aspirin (1)

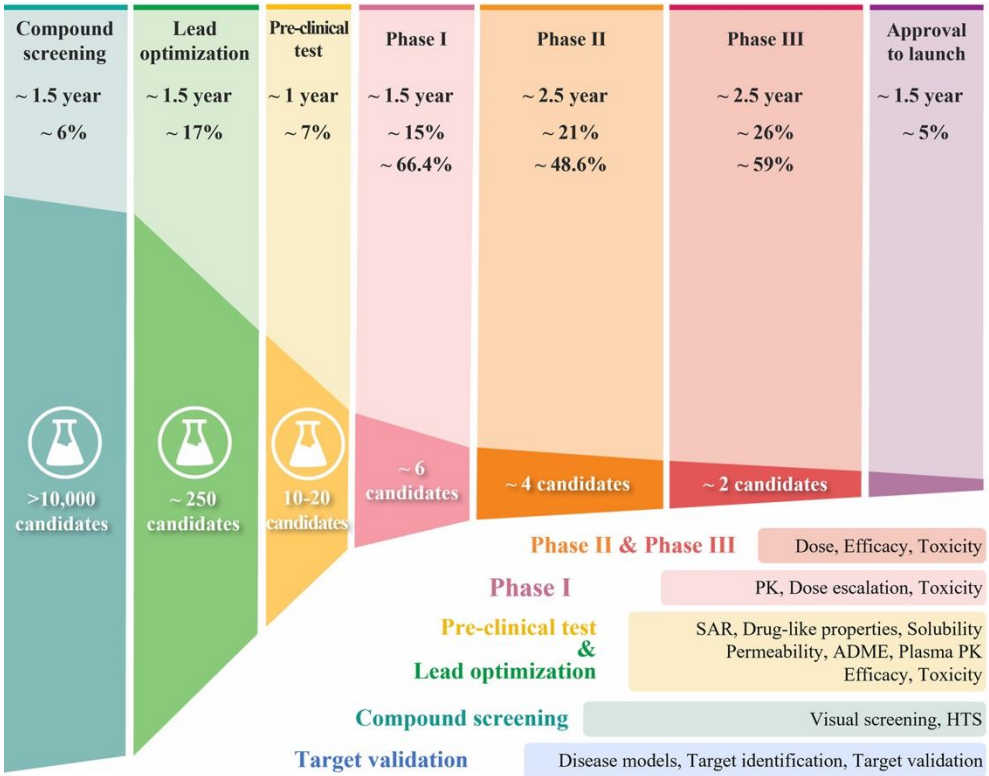


Ibuprofen (2)

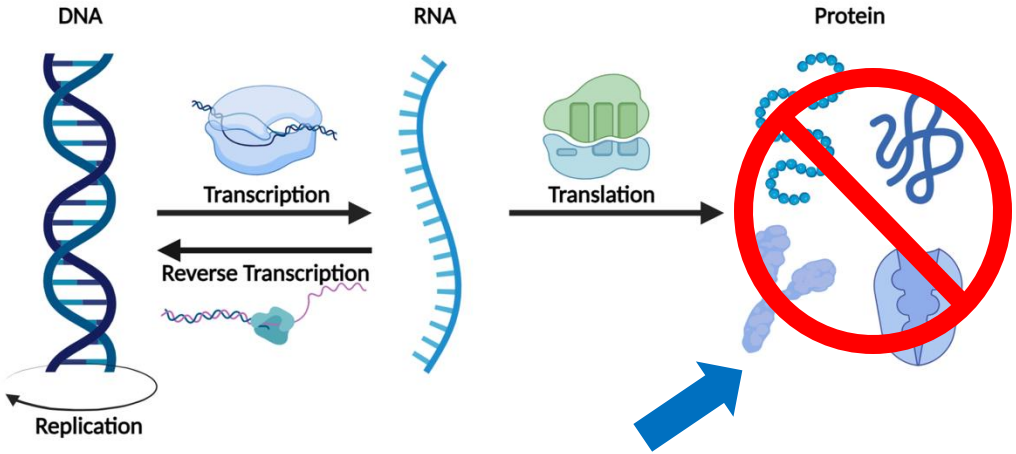
- Over 90% of the drugs on the market
- Low molecular weight
- Can pass through the cell membrane
- Taken as pills

Limitations of small molecules drugs

- Small molecule drug discovery relies on large-scale screening
- Time-consuming
- Expensive
- High failure rates
  - Lack of efficacy
  - Unexpected toxicity



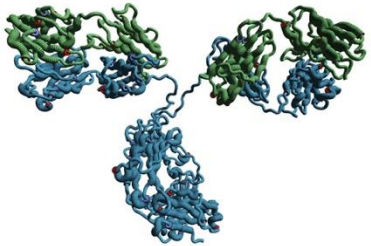
Biologics/therapeutic proteins as drugs



↑ Upregulation
↓ Downregulation
╳Misfolding/dysfunctional mutation

Biologics/Therapeutic proteins

Insulin
Humira (Adalimumab)
Keytruda (Pembrolizumab)
Opdivo (Nivolumab)
Eylea (Aflibercept)
Stelara (Ustekinumab)



- High specificity and affinity to the target molecule
- Higher efficacy and fewer adverse events
- Wider range of applicable targets

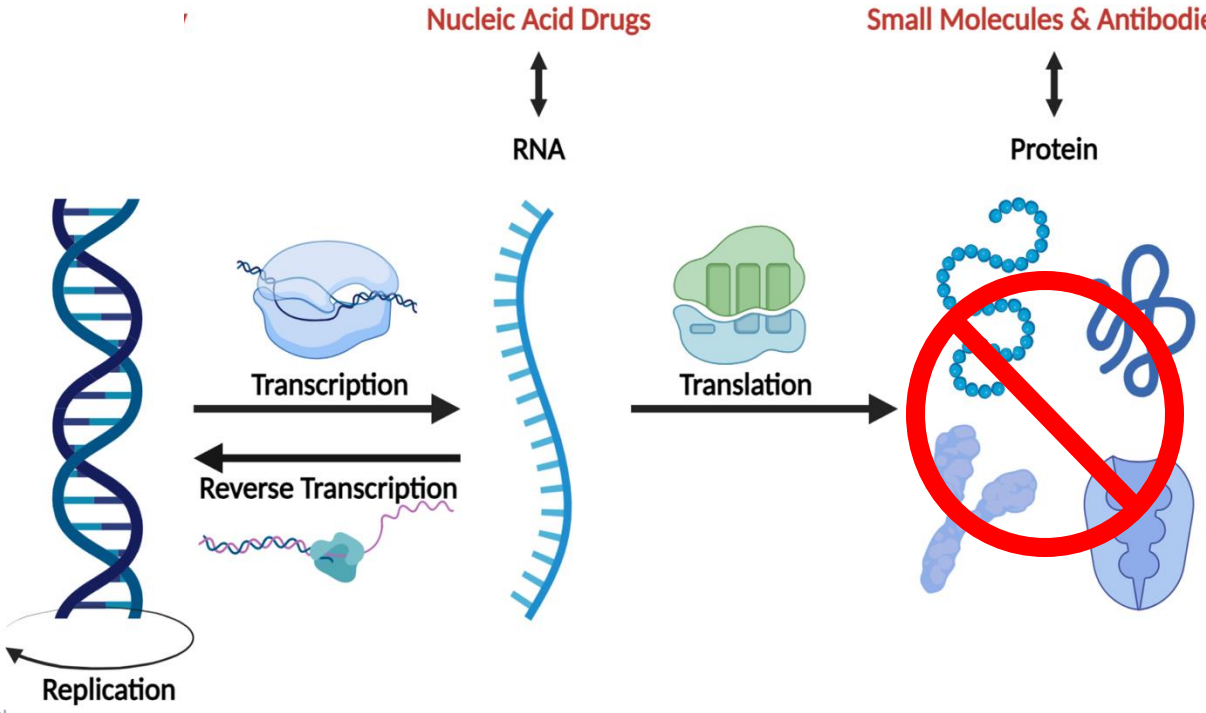
Limitations of biologics/therapeutic proteins

- Made by engineered cell lines in large bioreactors
- Expensive manufacturing processes with a large footprint
- Individualized optimization to stabilize and formulate proteins
- Difficulties in entering the cells
- Limited access to membrane-bound and intra-cellular proteins

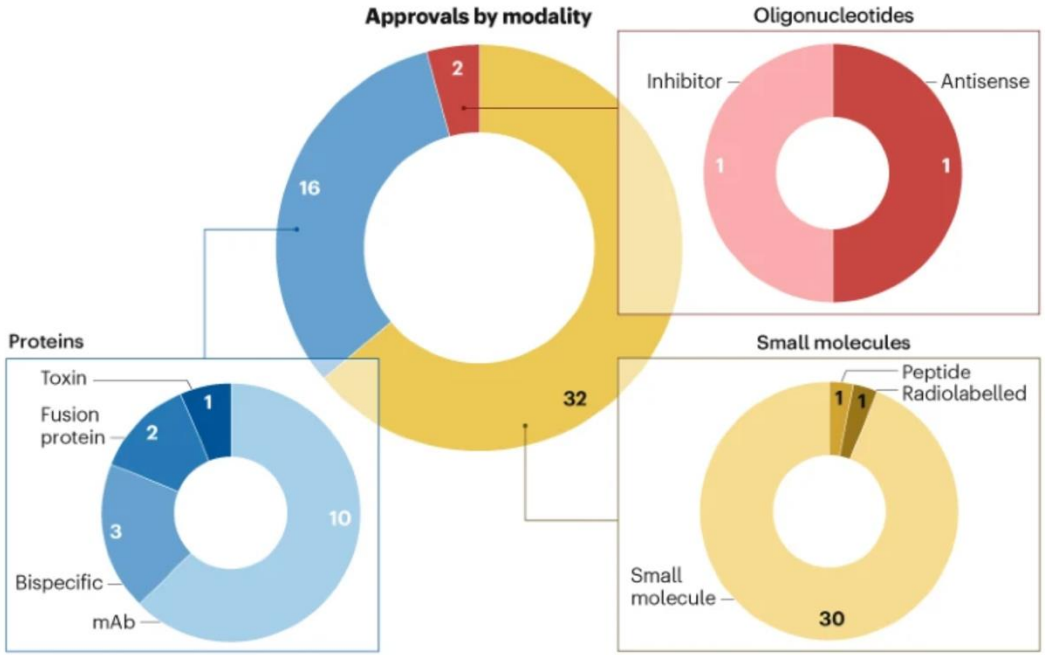
RNA as drugs

- Target mRNA, DNA, and protein
- Sequence complementarity

- Target proteins
- Needs specific binding pockets



New drugs approved in 2024 by modality



Advantages of RNA as therapeutics

1. <u>Targeting undruggable targets</u>
   - The ability to target any genetic components within the cell
   - Potential solution for undruggable targets using small molecules and biologics

Advantages of RNA as therapeutics

2. <u>Fast production and platform adaptability</u>
   - An established pipeline can be quickly adapted for similar applications

3. <u>Rare disease</u>
   - Lower cost for developing new therapies by adapting an established pipeline

Advantages of RNA as therapeutics

4. <u>Long-term effects</u>
   - Ongoing efforts in enhancing RNA stability and prolonging therapeutic effects
   - i.e. silencing effect of Inclisiran (siRNA targeting PCSK9) last more than six months with one dose

5. <u>No risk of genotoxicity</u>
   - Does not make changes or induce mutations at the DNA level

RNA therapeutics currently in clinic

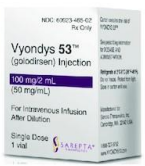
- As of now, Food and Drug Administration (FDA) approved
  - 13 Antisense oligonucleotide-based drugs
    - Fomivirsen, Mipomersen, Nusinersen, Eteplirsen, Inotersen, Volanesorsen, Golodirsen, Viltolarsen, Casimersen, Tofersen, Eplontersen, Imetelstat, Olezarsen
    - First ASO drugs approved in 1998
    - 12 drugs approved between 2013-2024
  - 6 siRNA-based drugs
    - Patisiran, Givosiran, Lumasiran, Inclisiran, Vutrisiran, Nedosiran
    - Approved between 2018-2024
  - 3 mRNA vaccines
    - Comirnaty (Pfizer-BioNTech COVID vaccine) and Spikevax (Moderna COVID vaccine)
    - mRESVIA (Moderna respiratory syncytial virus (RSV) vaccine)
    - Approved between 2020-2024

RNA therapeutics currently in clinic

ASO 1998
CMV Retinitis

Aptamer 2004
Age-Related Macular Degeneration (AMD)

ASO 2013
Familial Hypercholesterolemia



SSO 2016 **SSO 2019**
Duchenne Muscular Dystrophy

SSO 2016
Spinal Muscular Atrophy

siRNA 2023
primary hyperoxaluria

ASO 2023
Hereditary Transthyretin Amyloidosis With Polyneuropathy

Vaccine Adjuvant 2017
HBV

siRNA 2018
TTR Polyneuropathy

ASO 2018
TTR Polyneuropathy

waylivra®
(volanesorsen) injection 285 mg/1.5 mL

ASO 2019 EU
Familial Chylomicronemia Syndrome

siRNA 2019
AHP Porphyria

siRNA 2020
Primary Hyperoxaluria type 1

mRNA 2024
respiratory syncytial virus

ASO 2024
myelodysplastic syndromes

SSO 2020
Duchenne Muscular Dystrophy

SSO 2021

siRNA 2021
LDL-C reduction "siRNA statin"

siRNA 2022
TTR Polyneuropathy

mRNAs 2021/2022
Covid-19 Vaccines

ASO 2023
amyotrophic lateral sclerosis

ASO 2024
familial chylomicronemia syndrome

Created by Elaine Zhelan Chen

RNA Biology

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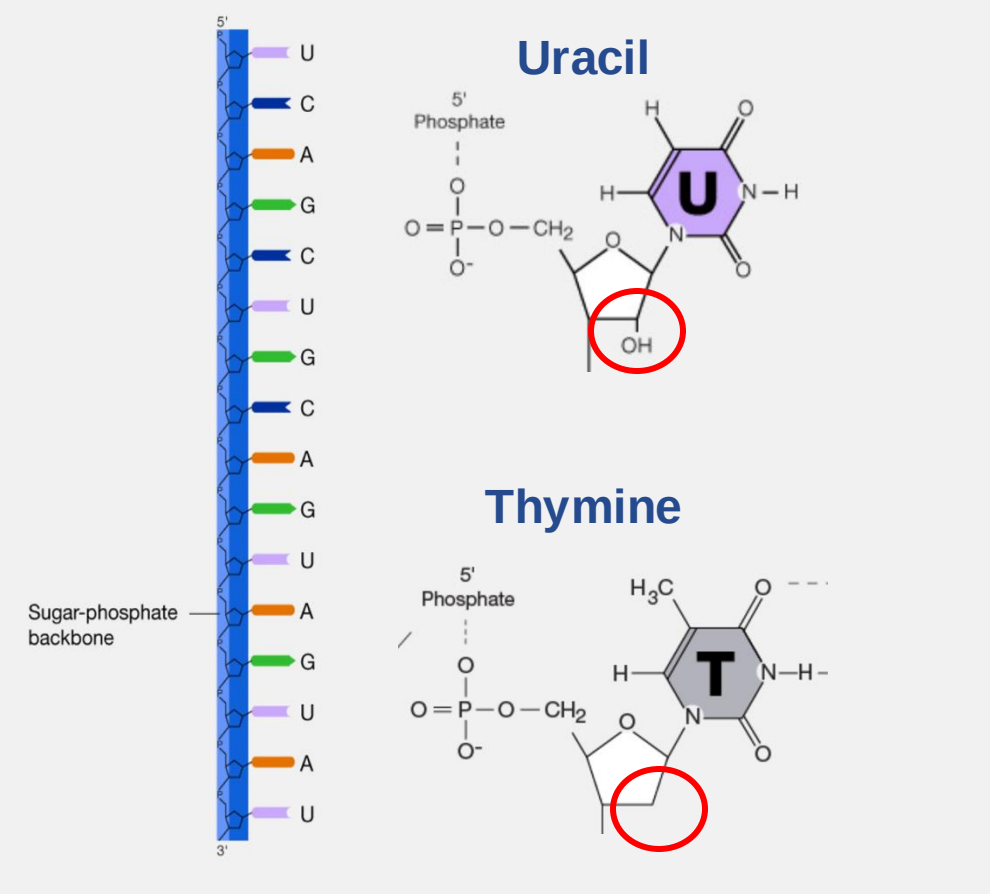
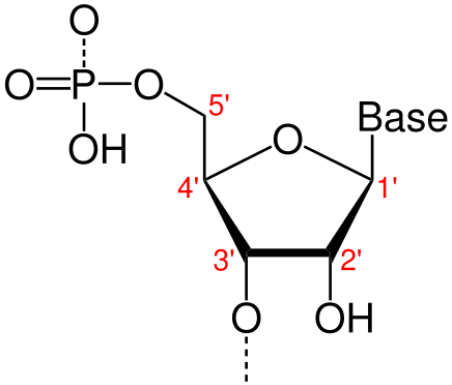
RNA

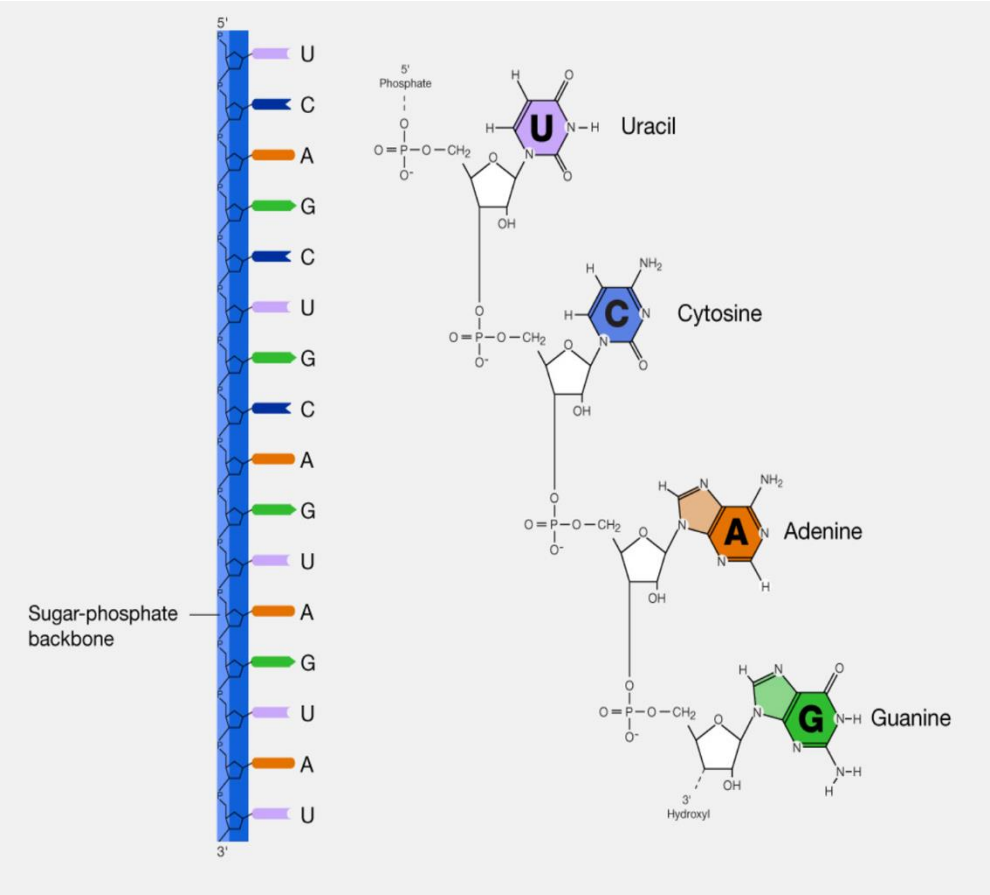
- Ribonucleic acid (RNA)
- Unlike DNA, RNA is most often single-stranded
- Backbone made of alternating phosphate groups and the sugar ribose, instead of deoxyribose found in DNA
- Strongly negatively charged because of the phosphate group



RNA

- Four bases: adenine (A), uracil (U), cytosine (C) or guanine (G)
- RNA is synthesized in vivo from 5’ to 3’
- A new nucleotide is added to the 3' hydroxyl (-OH) group via a phosphodiester bond





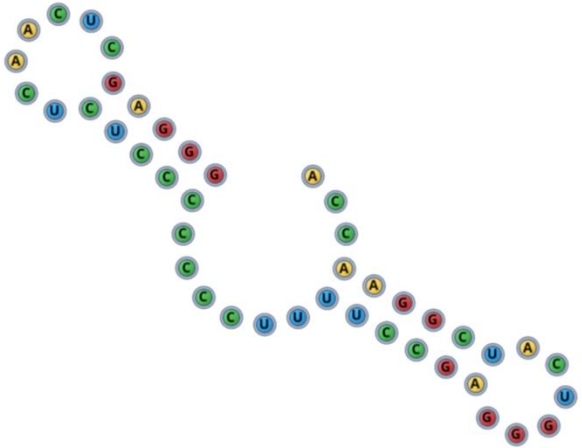
RNA 2D and 3D structures

Primary Structure:

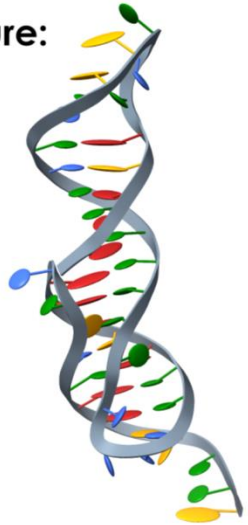
GGGAGCUCAACUCUCCCCCCCUUUUCCGAGGGUCAUCGGAACCA

.....((((((((....))))))))....................

Secondary Structure:

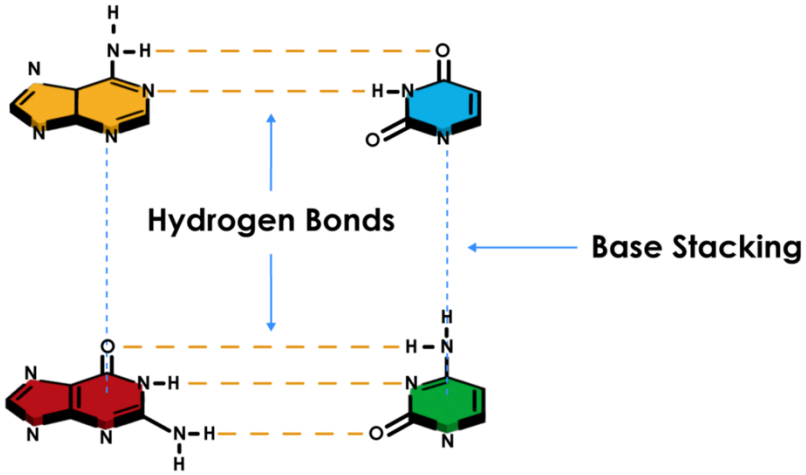


Tertiary Structure:



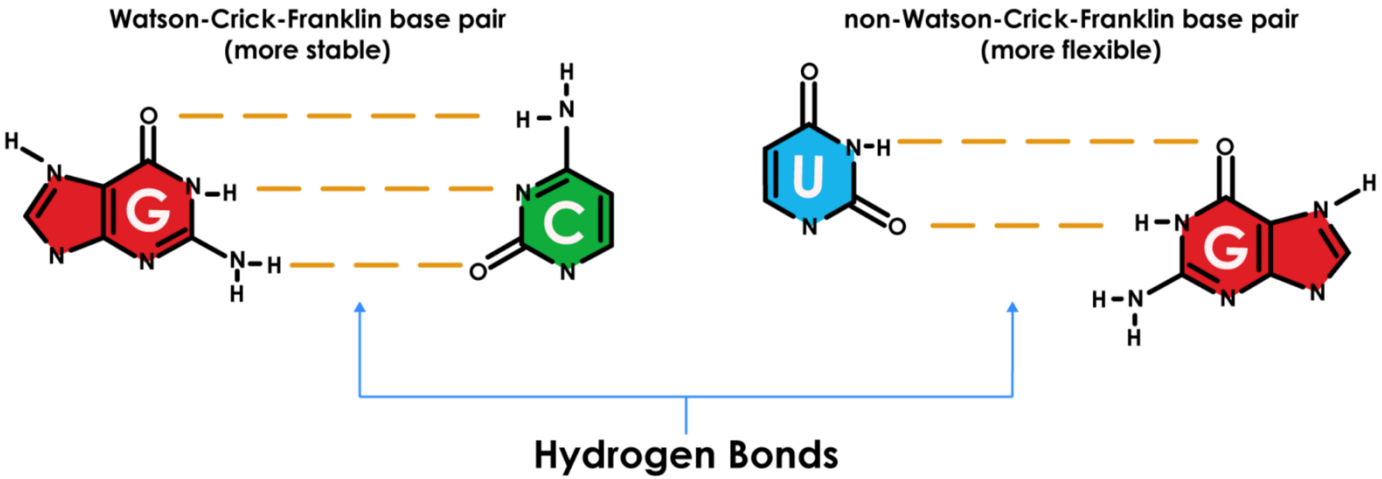
Even with the exact same sequence of A, U, C, G, RNA molecules can form different 3D structures

Hydrogen bonds and base stacking



Two RNA loops bound to each other by base stacking interactions and hydrogen bonds.

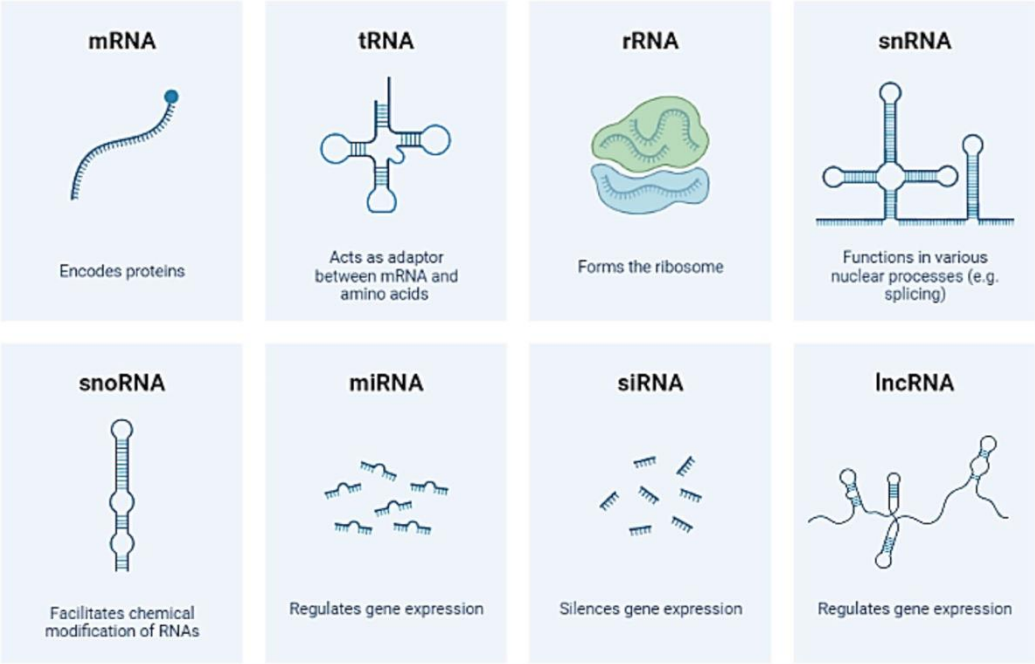
RNA base-paring



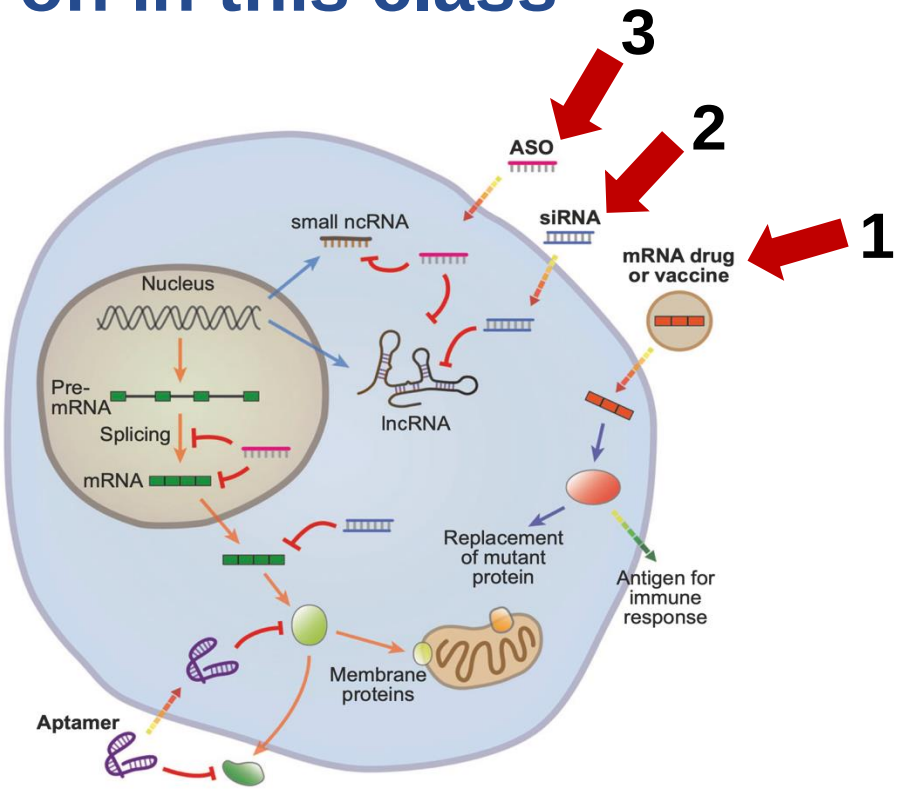
Different types of RNA

Coding RNA
- messenger RNA (mRNA)

Non-coding RNA
- ribosomal RNA (rRNA)
- transfer RNA (tRNA)
- small nuclear RNA (snRNA)
- microRNA (miRNA)
- small interfering RNA (siRNA)
- long non-coding RNA (lncRNA)
- small nucleolar RNA (snoRNA)



Types of RNA therapeutics that we will focus on in this class



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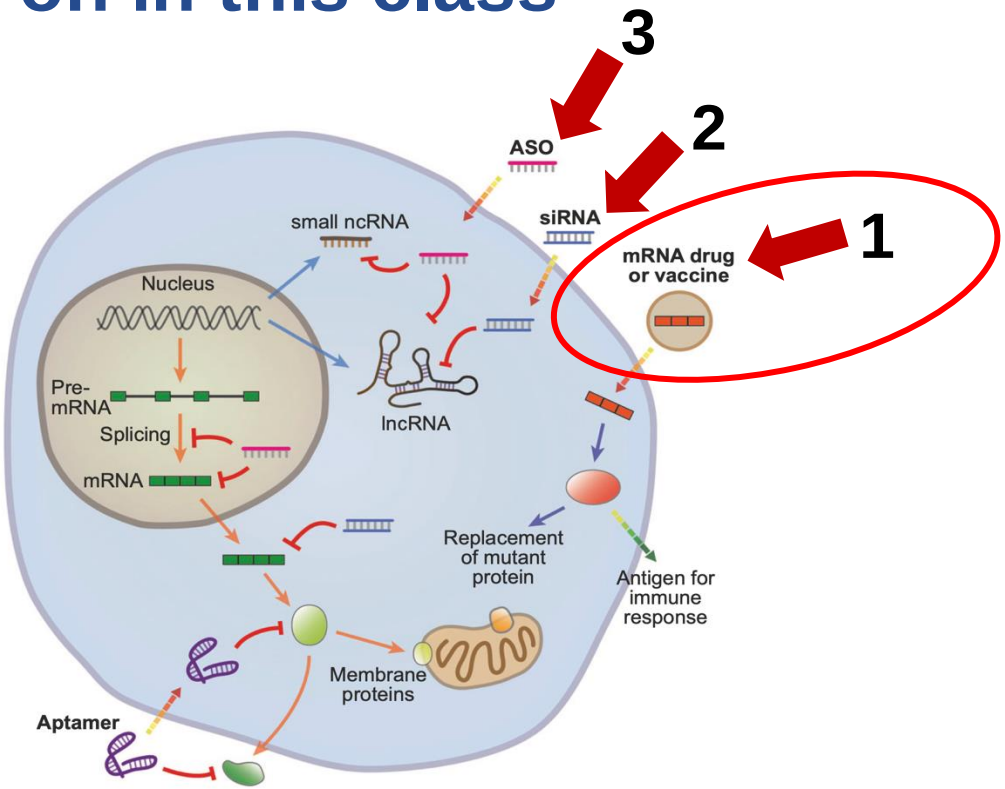
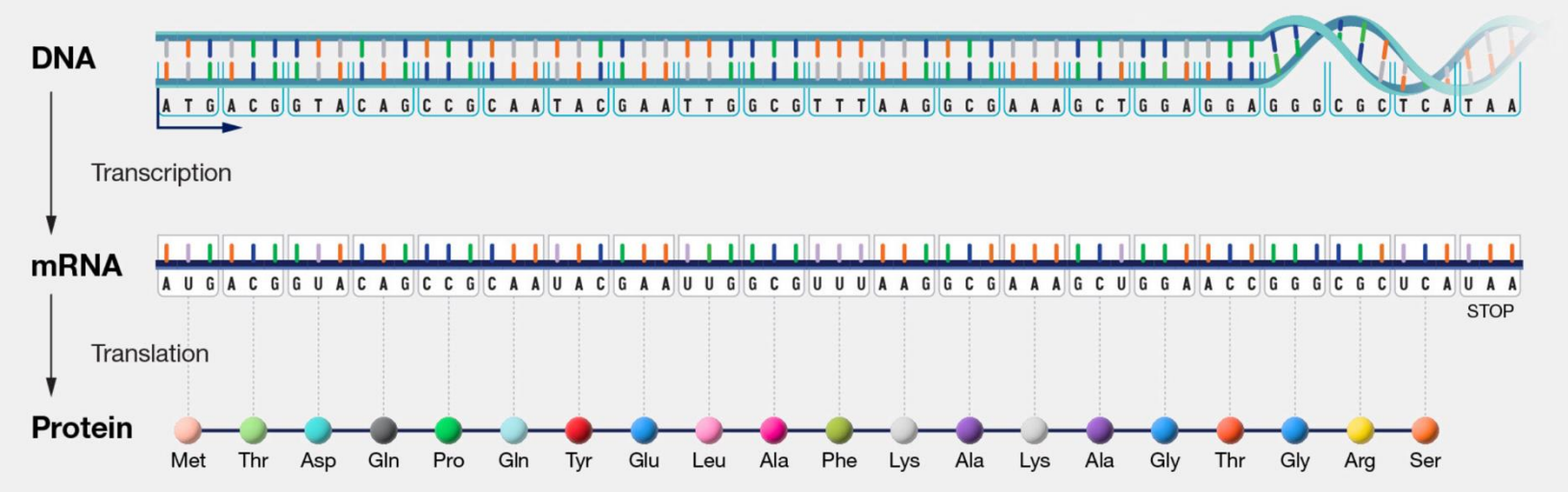
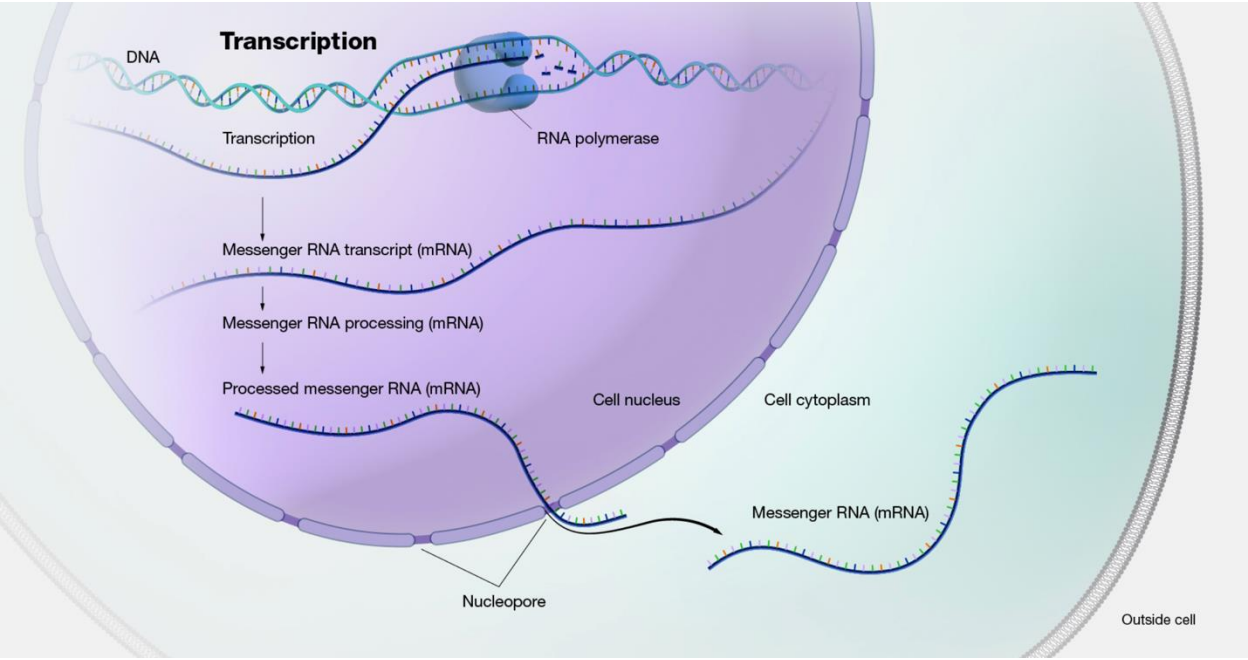


The flow of genetic information in eukaryotic cells



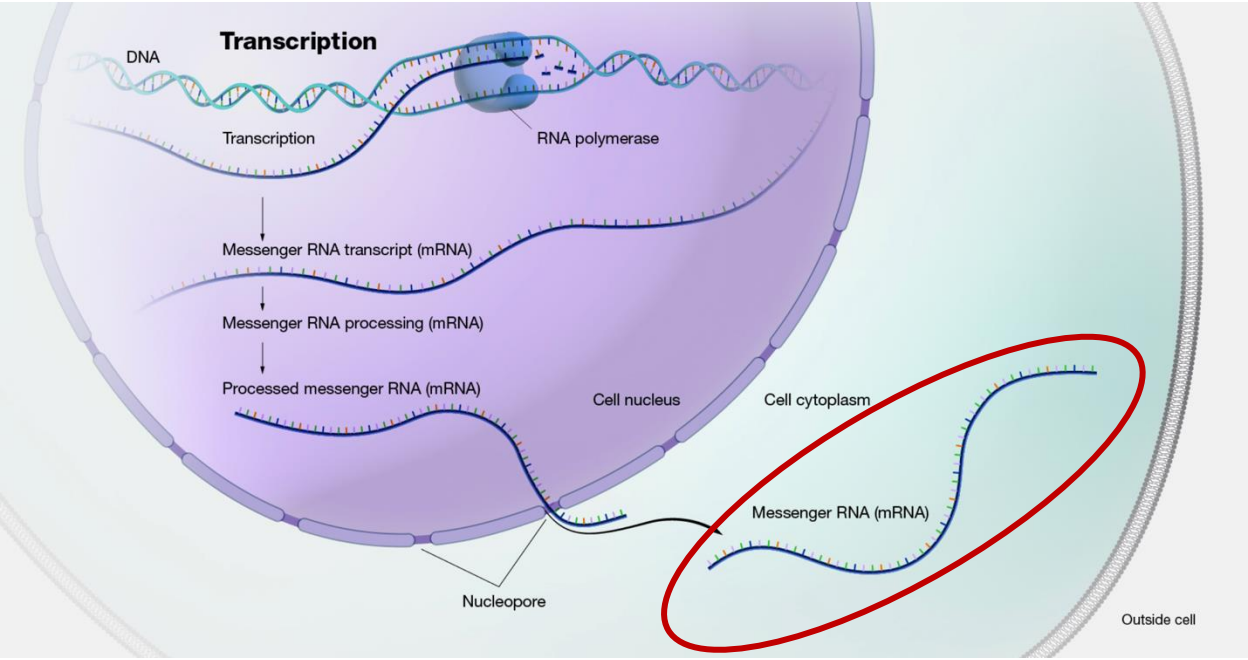
The flow of genetic information goes from DNA, to RNA, to protein, or directly from RNA to protein.

Transcription



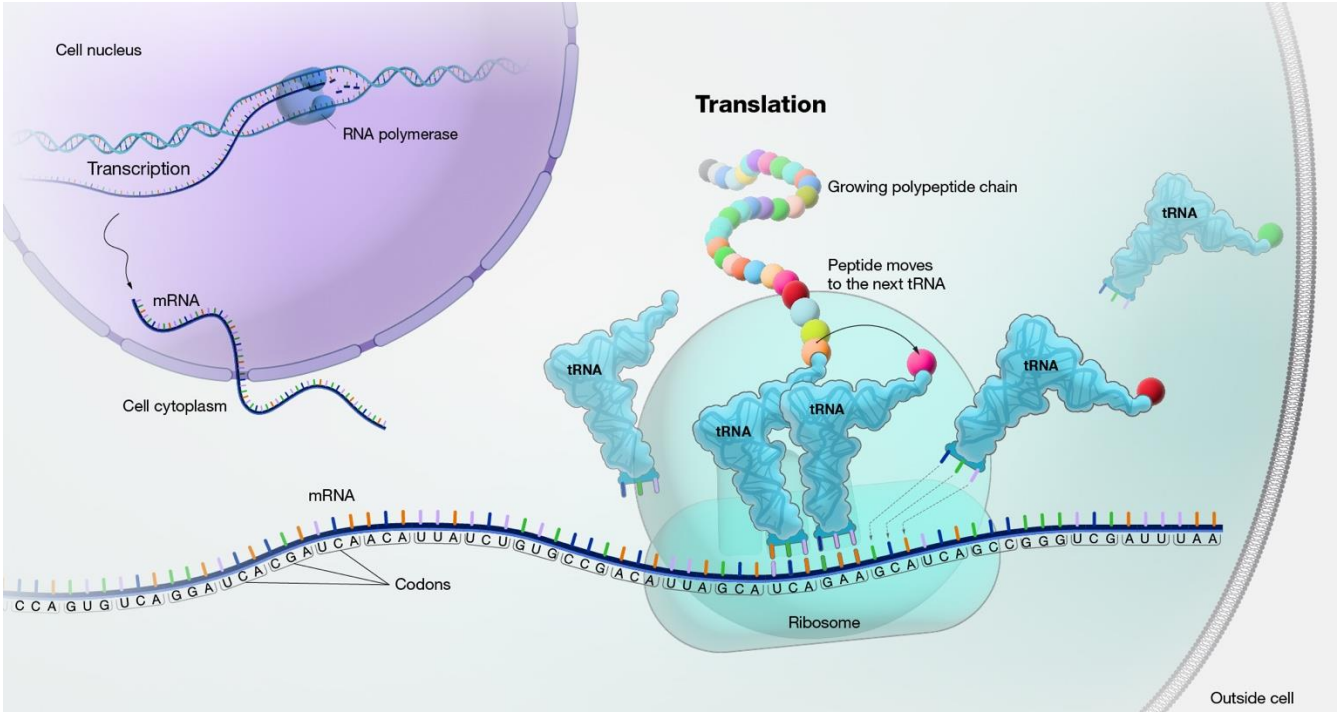
Transcription is a process of making an RNA copy of a gene's DNA sequence. This copy is the messenger RNA (mRNA).

Messenger RNA (mRNA)



mRNA moves from the nucleus into cytoplasm, where the protein-making machinery reads the mRNA sequence and translate each three-base codon into its corresponding amino acid

Translation



Translation takes place on ribosomes in the cell cytoplasm, where mRNA is read and translated into the string of amino acid chains that make up the synthesized protein.

Codons

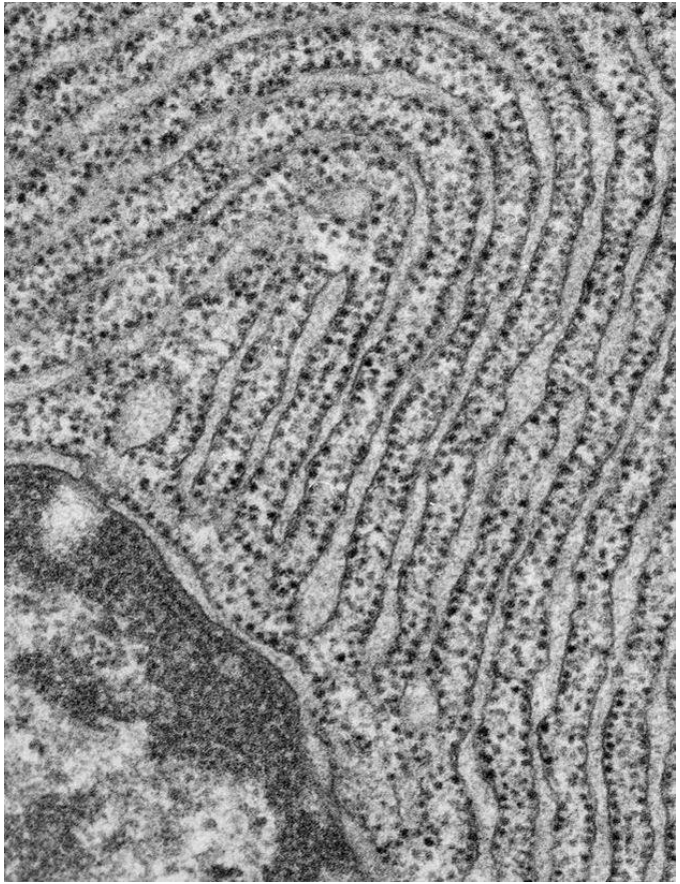
- 4 nucleotides
- 64 combinations
- Encodes for 20 different amino acids
- Within the 64 codons
  - AUG encodes for Methionine, the start codon signaling the START of translation
  - UAA, UGA, and UAG encodes for the STOP codon, signaling the end of translation
- Multiple codons can encode for the same amino acid
- Therefore, multiple RNA sequences can encode for the same protein

Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC } Phe UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Tyr **UAA Stop** **UAG Stop**	UGU, UGC } Cys **UGA Stop** UGG Trp	U C A G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U C A G
A	AUU, AUC, AUA } Ile **AUG Met**	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Asn AAA, AAG } Lys	AGU, AGC } Ser AGA, AGG } Arg	U C A G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U C A G

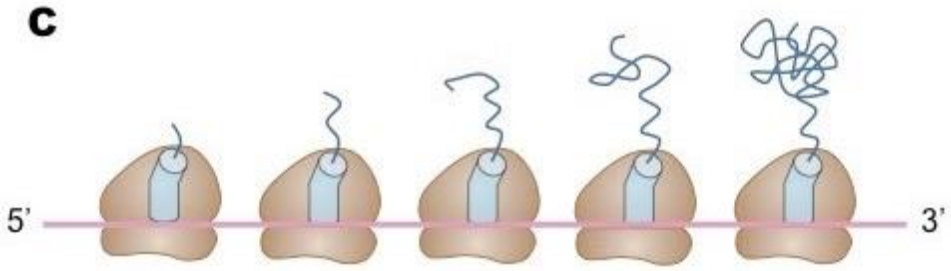
A more realistic view

- About 3 million ribosomes in a human cell
- Ribosomes can string together 60 amino acids per minute
- Each human protein take a few minutes to make



A more realistic view

- Each mRNA molecule can be translated hundreds to thousands of times
- A polysome is formed when one mRNA molecule is being translated by more than one ribosome



mRNA as therapeutics

mRNA as Therapeutics

- mRNA can code for any protein
- mRNA can be used for
  - Vaccines to protect against new pathogens
  - Immunogenic therapies to fight against cancer
  - Providing instructions to make missing proteins to restore function
  - And more

The way mRNA therapeutics work is enabling the human body to make its own medicines in the form of protein

Emergence of mRNA therapeutics

mRNA

- mRNA is discovered in 1961
- Expression of exogenous mRNA in mice described in 1990
- First proof of concept study for mRNA vaccine against Influenza in 1993
- First design of mRNA vaccine to treat cancer in 1995
- Clinical trial on mRNA vaccine to treat metastatic melanoma in 2008
- Clinical trial on mRNA vaccine to treat rabies, an infectious disease, in 2013
- Emergency approval of SARS-CoV-2 mRNA vaccine in 2020
- mRNA RSV vaccine approved in 2024

mRNA challenges - delivery

mRNA delivery

- mRNA needs to reach the cell cytosol to be translated into protein to achieve therapeutic effects
- mRNA cannot get into the cells on their own
  - RNA is strongly negatively charged
- mRNA is easily degraded
  - Rapidly degraded by extracellular RNases
- mRNA itself lacks cell-type targeting capabilities

mRNA delivery

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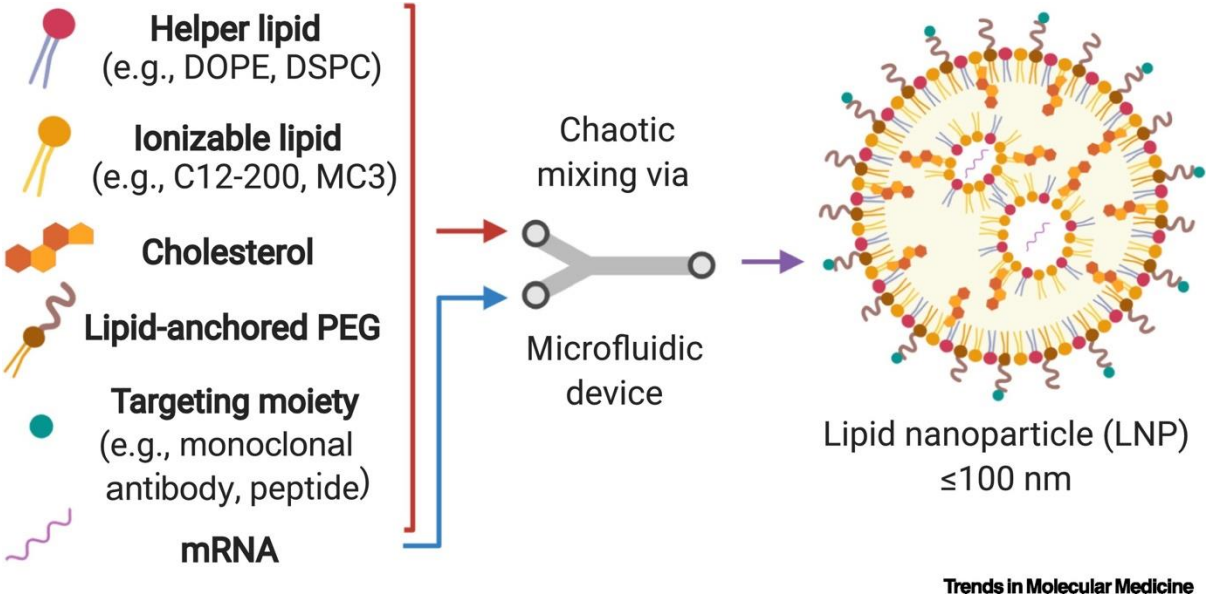
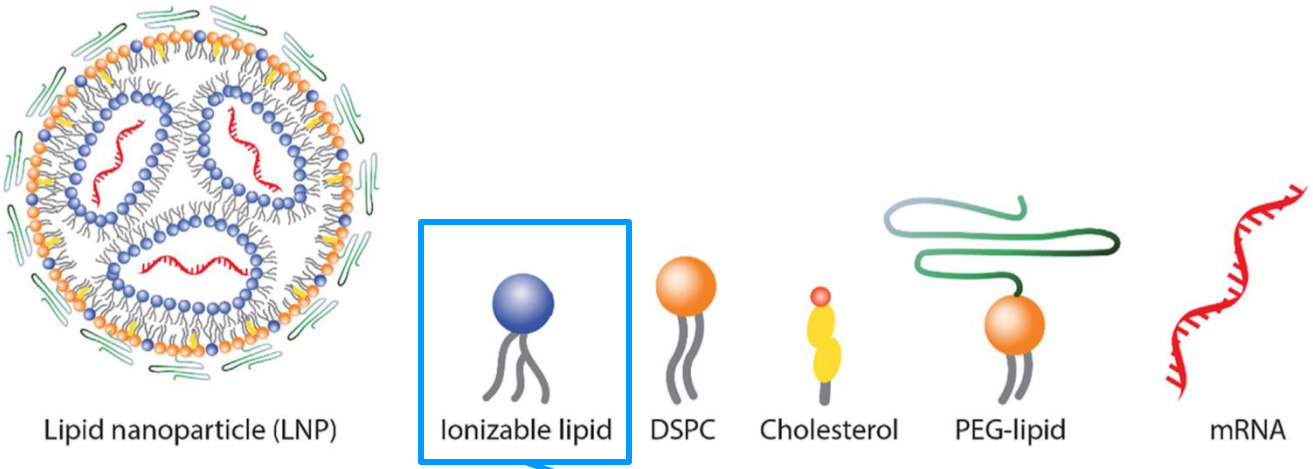
mRNA requires a delivery system that protects the nucleic acid from degradation, allows cellular uptake, and helps targeting a specific cell type

Lipid Nanoparticles (LNP)



Lipid nanoparticles (LNPs) have emerged as a highly efficient delivery system for exogenous mRNA in vivo

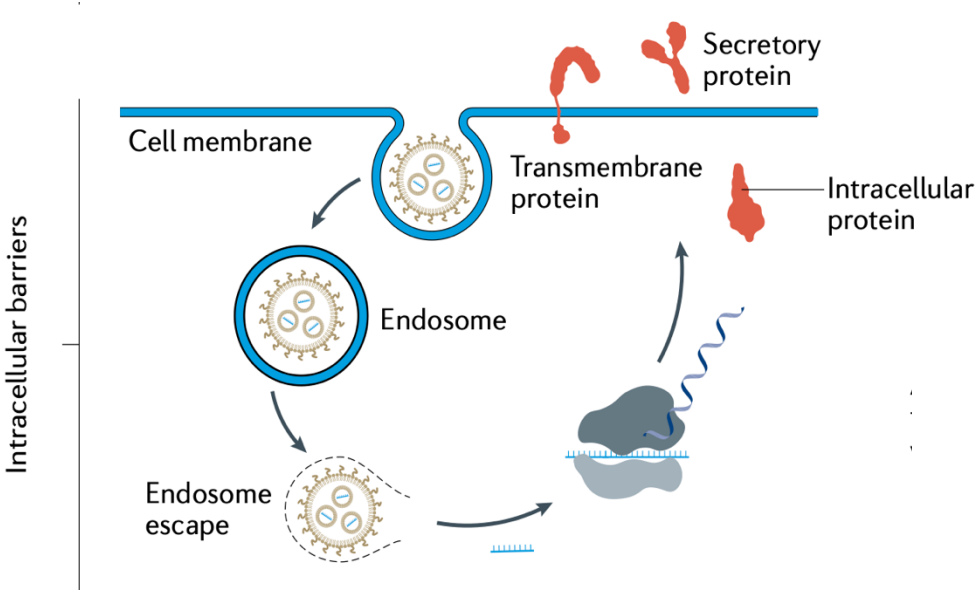
Ionizable lipid



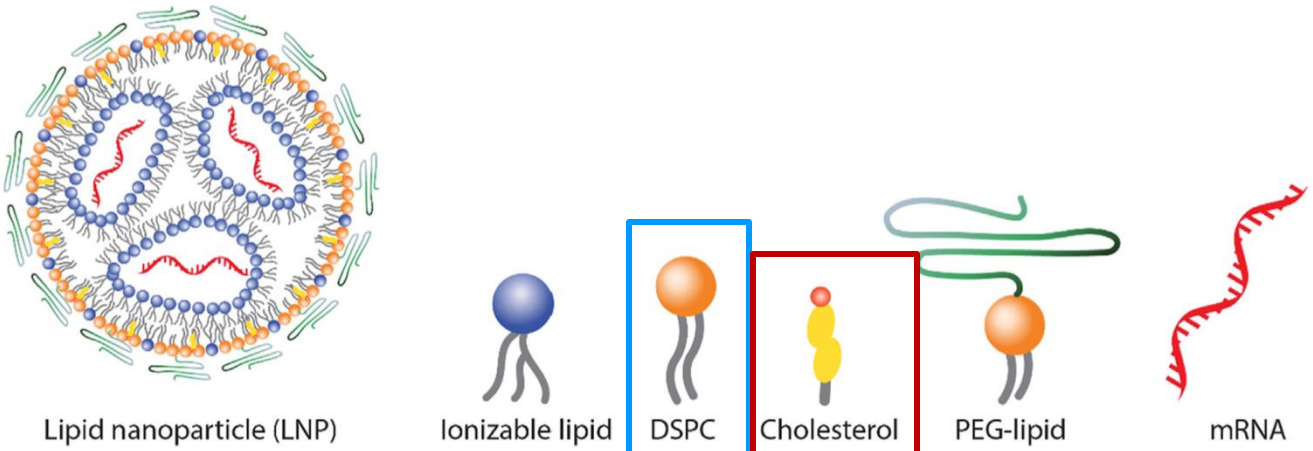
- Also referred to as cationic lipid
- Facilitate mRNA encapsulation in LNPs
- Positively charged under acidic condition, uncharged under physiological pH (around neutral pH) to avoid cytotoxicity
- Mediate endosomal membrane disruption to enable mRNA release to the cytosol

LNP-mRNA after entering the cell

- LNPs are first internalized by the cells
- Subsequently, LNP are usually trapped in endosomal compartments
- Only a small amount of lipid nanoparticles may be able to escape from the endosome
- Endosomal escape is crucial for effective mRNA delivery
- Ionizable lipid can electrostatically bind to the endosomal membrane which may allow LNPs to destabilize it and facilitate escape



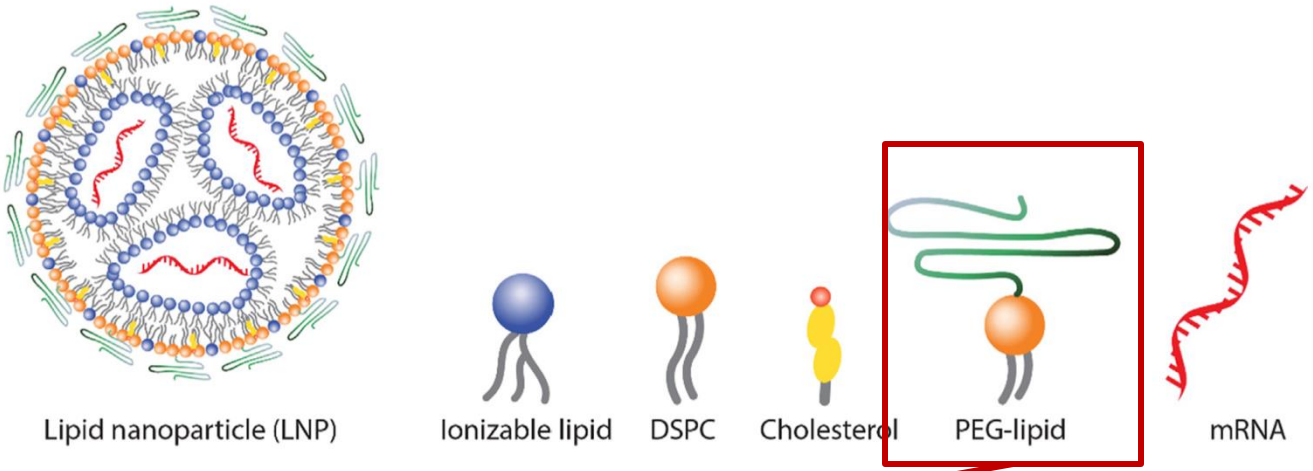
Helper lipids



DSPC
- Enhance stability
- Assist forming of stable lipid bilayer

Cholesterol
- Enhance stability of the lipid bilayer

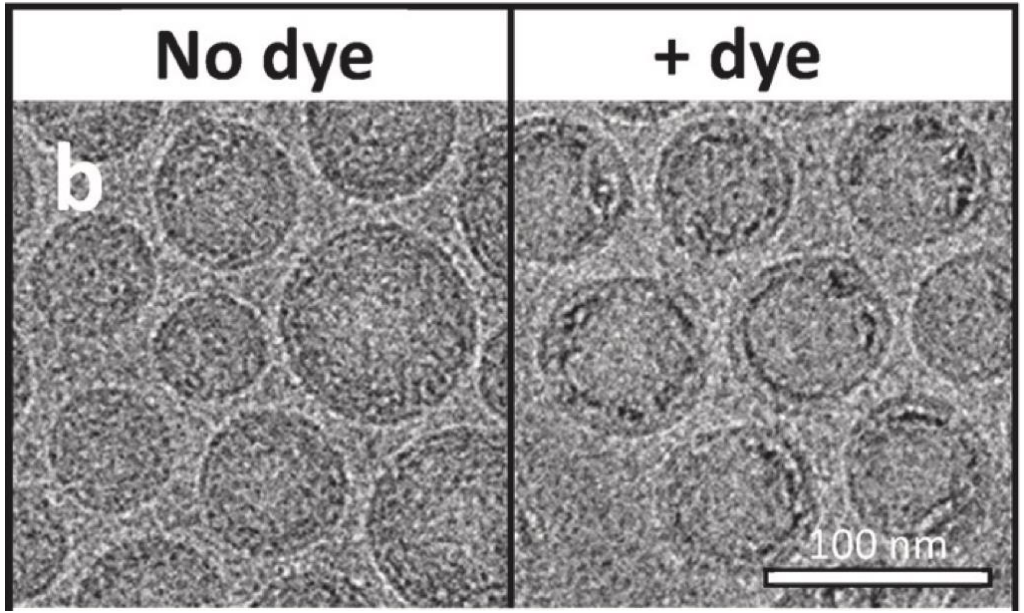
PEG-lipid (Polyethylene glycol lipid)



PEG-lipid

- Prevent LNP aggregation
- High stability and stealth-like properties
- Low cellular uptake and poor endosomal escape

mRNA in LNP



Each LNP contains 4-6 mRNA molecules

Final Presentation

Final presentation

- 4 drugs in different stages of clinical trials
  - mRNA enzyme replacement therapy
  - mRNA cancer vaccine
  - siRNA Plozasiran for hyperlipidemia
  - siRNA futisiran for hemophilia

- 20 min in total + 5 min Q&A

- Google sheet for sign up
  - Team formed before lecture on Wednesday, Jan 8th, 2025

Final presentation

<u>Handout</u>

Objectives:

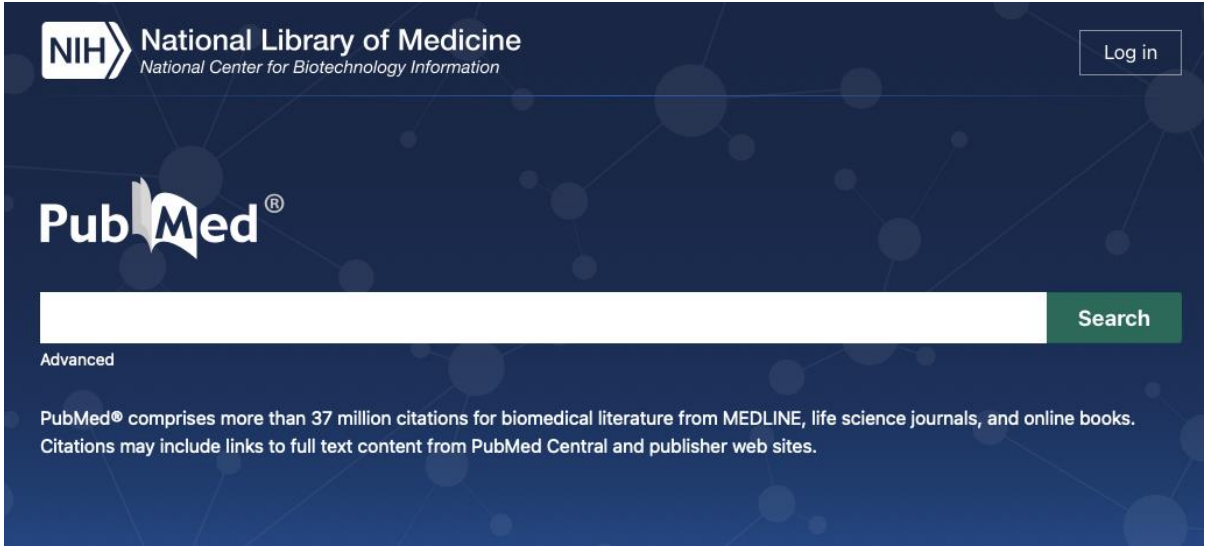
- **Understand the drug's scientific and clinical background**: What is the disease of interest? What is mechanism of action of the drug? How does this drug compare to existing treatments?
- **Analyze early proof of concept data**: Summarize early proof of concept studies. What is the key evidence showing that the drug is worth pursuing?
- **Analyze clinical trial data**: Summarize the latest clinical trial results.
- **Address safety and efficacy**: Discuss the drug’s safety profile, side effects, and overall efficacy in clinical trials.

Final presentation

Project package:
Item #1: A paper of early proof of concept data
Item #2: A paper of the latest clinical trial
Item #3: A plain language summary, commentary article, or public slide deck. These are digested information to help you grasp the main talking points.

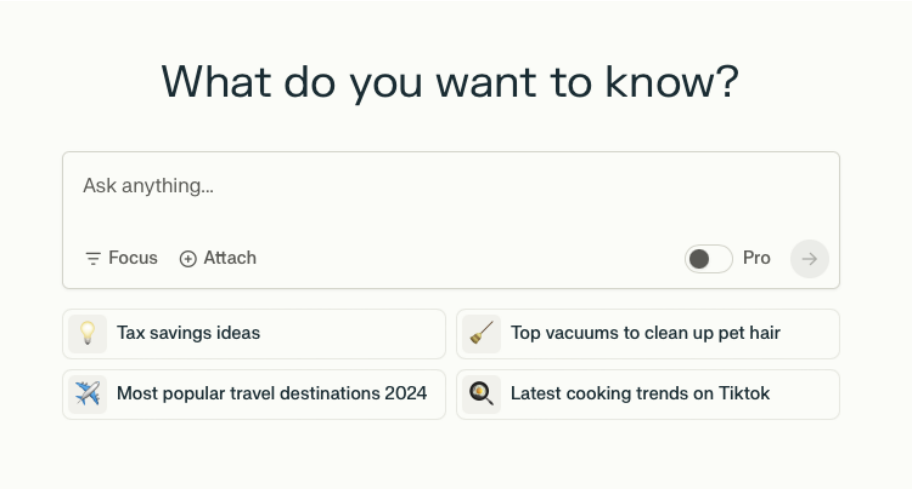
Literature search

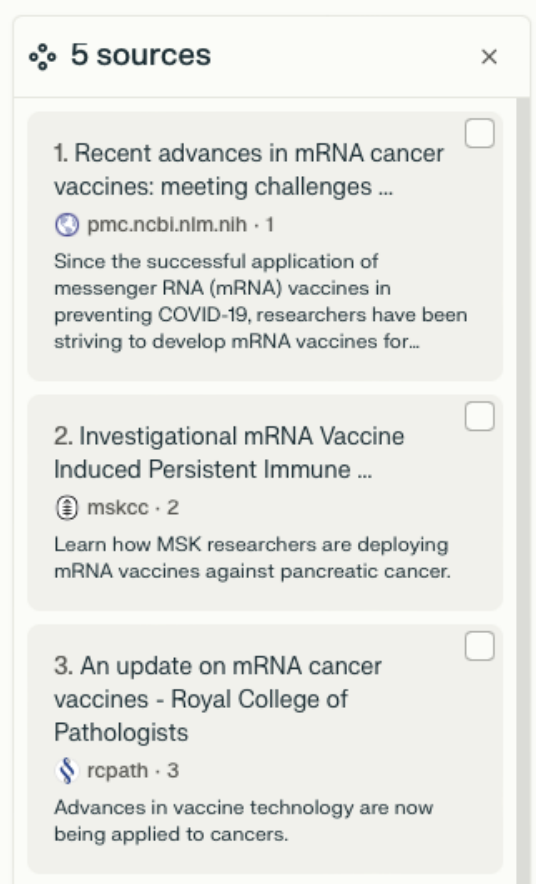
- Websites
  - Google
  - PubMed (https://pubmed.ncbi.nlm.nih.gov)
    - For searching for original scientific literatures



Supplementary tools

- ChatGPT
- Perplexity (https://www.perplexity.ai)
  - for quick and brief explanations with potentially useful references





Components of a original research paper

- Abstract – high-level summary of a paper
- Introduction – background, rationale, and major findings of a paper
- Results – in-depth explanations of data and figures
- Discussion - analysis and interpretation of findings, conclusion and next steps
- Methods – technical details of how experiments are carried out
- Reference
- Supplemental data – data discussed but not shown in the paper

Other sources of information

- Clinical trials (https://clinicaltrials.gov)
- FDA (https://www.fda.gov/drugs)
- Patents (https://patents.google.com)

Problem set #1

- Available after class
- **Due on Wednesday, Jan 8th, 5PM EST**

Attendance